

Chest X-Ray Pneumonia Screening

MEDIMG FS26 — Decision Support in Medical Imaging

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Binary triage: **NORMAL vs PNEUMONIA** · Reproducible repo on GitHub

Motivation

- **Clinical need:** prioritize likely pneumonia cases in busy workflows
- **Our role:** decision **support** — not autonomous diagnosis
- **Success criteria:** ranking quality (AUC) + transparent error types (FP/FN) at a stated threshold

Dataset & Protocol

Item	Choice
Data	<code>hf-vision/chest-xray-pneumonia</code> (CC BY 4.0)
Train	~5.2k (HF train)
Validation	10% stratified hold-out from train (HF val $\approx 16 \rightarrow$ too small)
Test	624 HF test images — never used for training
Input	224×224 RGB, ImageNet normalization

Dataset (quick snapshot)

- **Task labels:** NORMAL vs PNEUMONIA
- **Test set label count:** 234 NORMAL · 390 PNEUMONIA
- **Why this dataset:** small, public, easy to reproduce end-to-end; good for studying baselines + ablations
- **Caveat:** single dataset → limited external validity (no hospital/site shift tested)

Evaluation

- **ROC-AUC** — threshold-free ranking
- **Accuracy & F1** — at fixed threshold **0.5** (not tuned for deployment)
- **Confusion matrix** — false positives on normals vs false negatives on pneumonia
- **Robustness**: seeds 0–2 for main models; ablations: aug, loss, 25% train data

Baselines (test set)

Method	AUC	Acc	F1
Random (train prevalence)	0.52	0.58	0.69
Majority class	0.50	0.63	0.77
Frozen ResNet18 + logistic regression	0.95	0.84	0.89

Takeaway: strong pretrained features already solve most of the task.

Model 1 — ResNet18 + cross-entropy

- ImageNet **ResNet18**, fully fine-tuned
- Loss: **cross-entropy**
- Main variant: **augmentation on** (flip, rotation, color jitter, **RandomErasing**)

Model 2 — ResNet18 + focal loss

- Same **ResNet18** backbone (fine-tuned)
- **Focal loss** ($\gamma = 2$) instead of CE — focus on hard examples
- Trained with augmentation on (same protocol as Model 1 aug)

Results — fine-tuned vs baseline (test AUC)

Model	AUC (seed 0)	AUC (seeds 0–2 mean)
Frozen ResNet18 + LR	0.949	—
CE, no aug	0.935	—
CE, aug on	0.960	0.954
Focal, aug on	0.957	0.953

Interpretation: gains over frozen LR are **small (~0.01–0.02 AUC)** — useful but not dramatic.

Results — threshold 0.5 matters (234 normals on test)

False positives = normal scans called pneumonia (cm_{01} / 234):

Run	FP count	Rate on normals
Frozen LR	94	40%
CE, no aug	108	46%
CE, aug (seed 0)	140	60%
CE, aug (seed 1)	72	31%
CE, aug (seed 2)	70	30%

Interpretation: higher AUC does **not** always mean fewer FPs at 0.5; aug changed **score calibration**, not a clear win on all metrics.

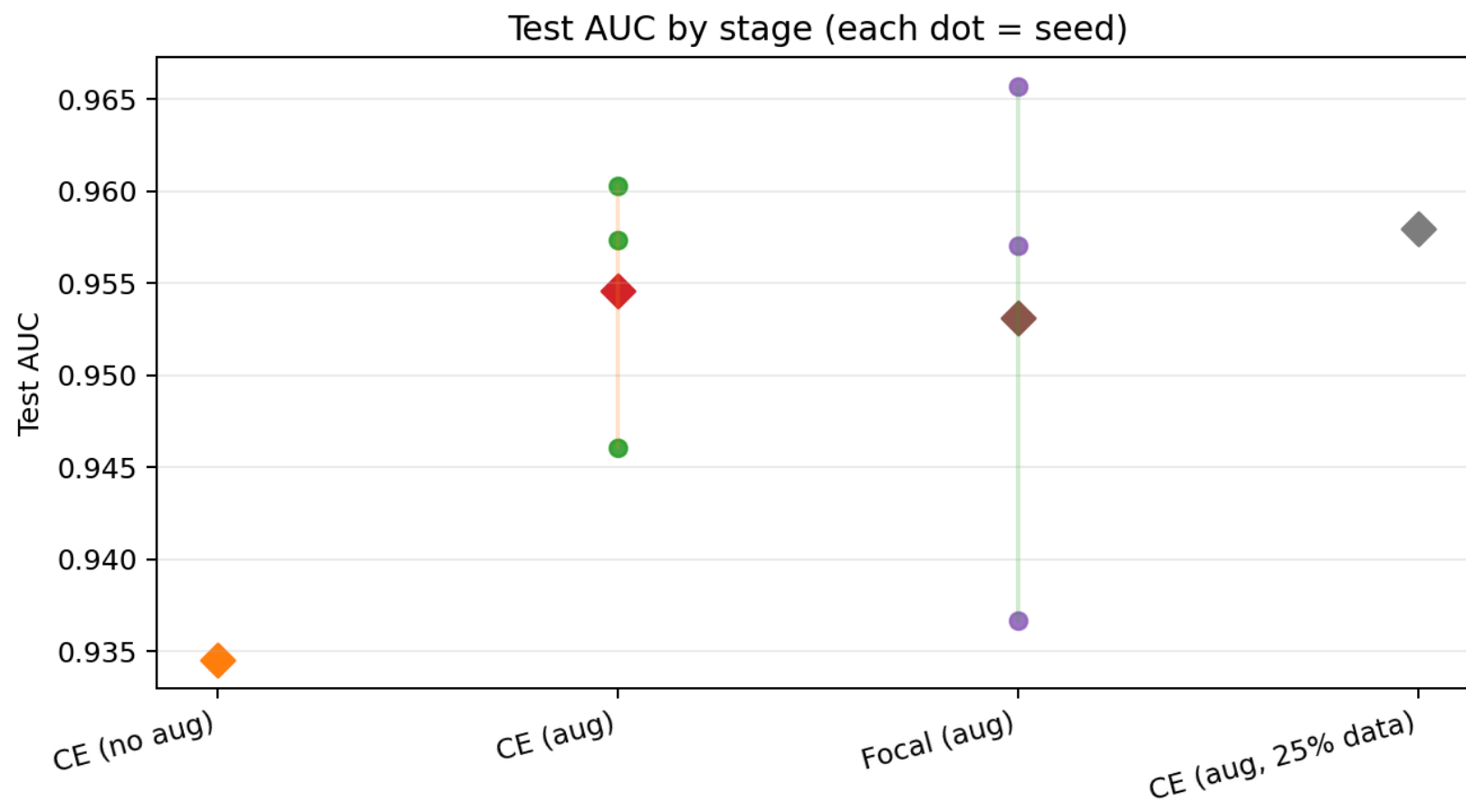
Ablations (seed 0, test)

Ablation	AUC	Acc	F1
Aug off (CE)	0.935	0.82	0.87
Aug on (CE)	0.960	0.78	0.85
Focal vs CE (aug on)	0.957 vs 0.960	~0.81 vs ~0.78	~0.87 vs ~0.85
25% train data (CE, aug)	0.958	0.85	0.89

25% run: **single seed** — illustrative only, not a strong claim.

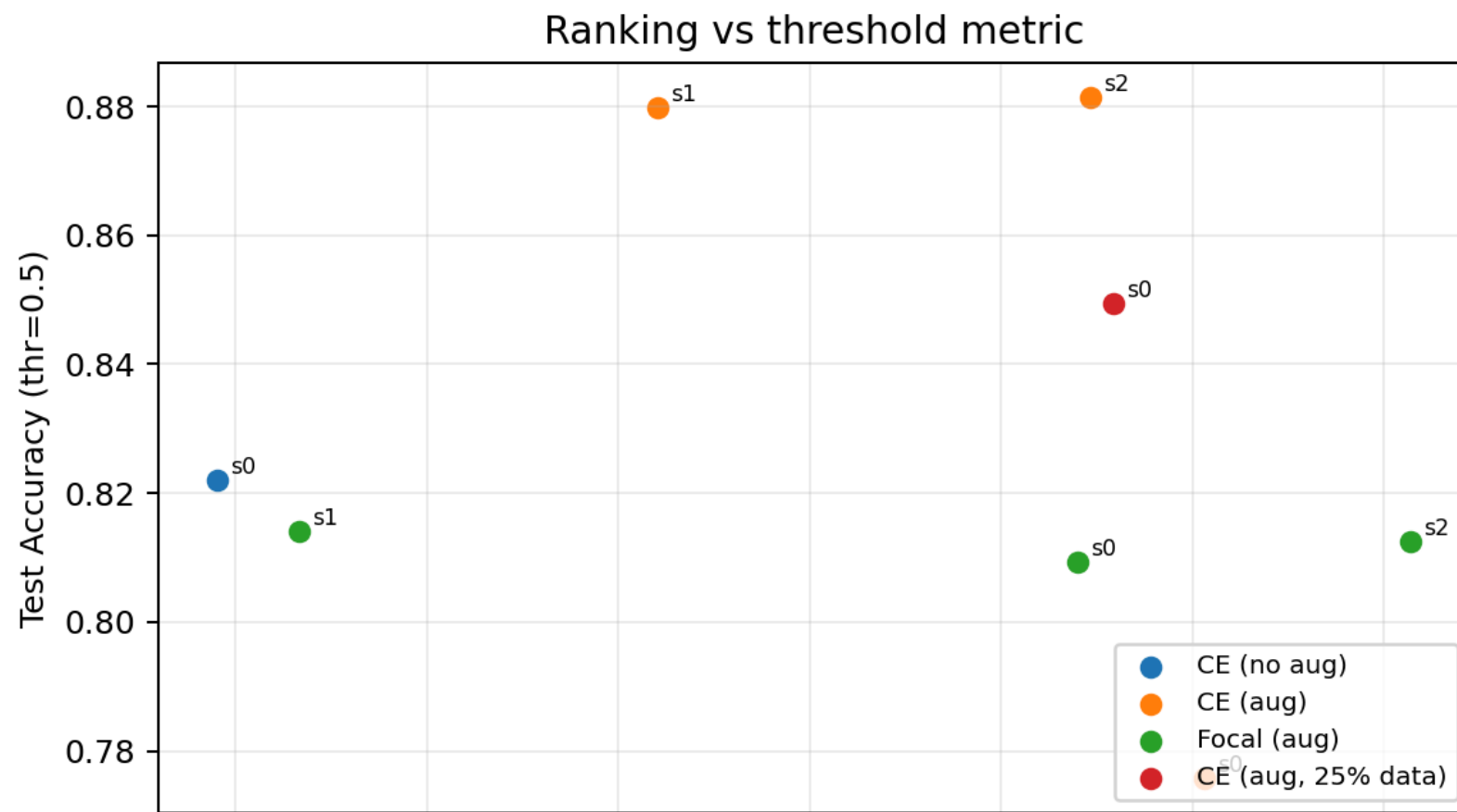
Summary plot — AUC across stages (seeds)

- AUC is **high** across runs; differences are **incremental**.
- CE-aug and focal-aug show **moderate** seed-to-seed variance.



Summary plot — AUC vs accuracy (thr=0.5)

- AUC measures **ranking**, accuracy depends on the **threshold**.
- Explains why some runs look “worse” at 0.5 even with strong AUC.



Qualitative analysis (Grad-CAM)

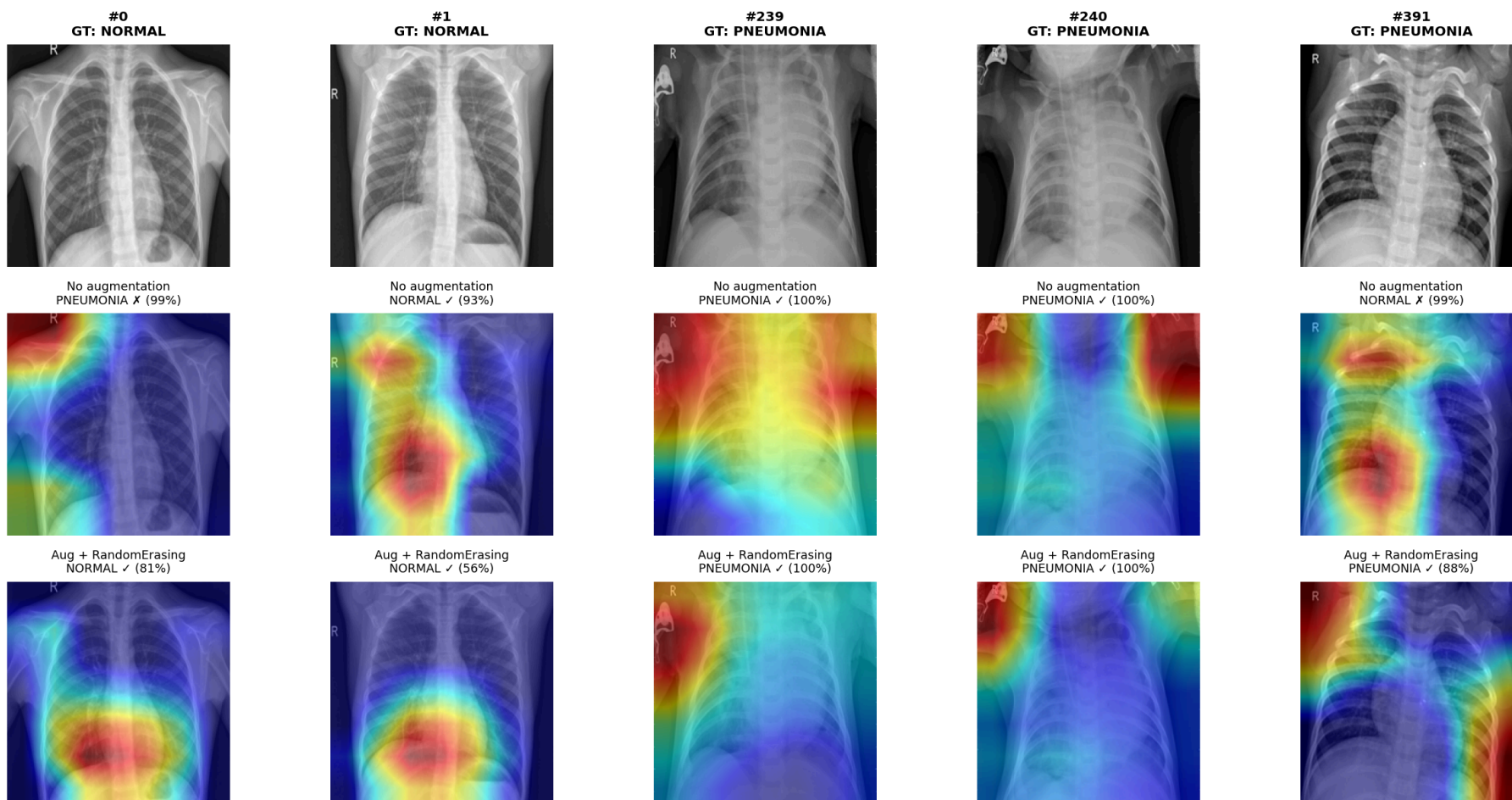
Goal: inspect *where* models look (explainability), not prove augmentation “fixes” attention.

Protocol:

- Fixed test indices → `configs/gradcam_manifest.json` (TP / TN / FP / FN examples)
- Grad-CAM targets **each model's predicted class**
- Compare **aug off vs aug on** (same images): `results/gradcam/aug_off_vs_on.png`

Qualitative — Grad-CAM (aug off vs on)

Grad-CAM for each model's predicted class



Qualitative — what we observed

- Maps often highlight **lung fields and borders** — not always a focal consolidation
- **Aug off vs aug on:** on several cases, CAM looks **more central / anatomical** (less “random” edge focus)
- **Quantitative** changes (AUC, FP rate at 0.5) were **larger than visual** differences
- **Honest conclusion:** saliency is **illustrative**; needs external validation + better methods for clinical trust

Limitations & ethics

- Single public dataset · risk of **shortcuts** (view, edges, equipment)
- **No threshold tuning** on validation for clinical operating points
- **No external site** validation · high AUC $\neq$ safe deployment
- **Automation bias** if users over-trust scores
- **False negatives** remain clinically critical — report FN counts, not only AUC

Reproducibility

```
bash scripts/setup.sh && source .venv/bin/activate  
bash scripts/reproduce.sh all  
SEEDS="0 1 2" bash scripts/reproduce.sh robustness
```

Artifacts: `results/test_summary.csv` , per-run plots under `results/<stage>/seed_*/` ,
checkpoints in `checkpoints/`

Conclusion

1. Task is **feasible**; frozen ImageNet features are already strong (**AUC $\approx$ 0.95**).
2. Fine-tuning + aug gives **incremental** ranking gains; **not** a large leap.
3. **Focal loss** did not clearly beat CE in our runs.
4. **Qualitative Grad-CAM** shows **mixed** behavior — several cases look **more central/anatomical** with augmentation, but this is not a replacement for quantitative evaluation.
5. **Future work**: threshold tuning, external validation, richer pathology labels (e.g. multi-label CXR).

Thank you

Repo: `MEDIMG-FS26-Chest-Pathology-Detection`

Questions?